COURSE INFORMATION

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School/Faculty:	**Software Engineering/Computing**		
Program name:	**Bachelor in Software Engineering with Honours**		
Course code:	**SCSE2103**	**Academic Session/Semester:**	**2025/26 - 1**
Course name:	**Data Structures and Algorithms**	**Pre/co requisite (course name and code, if applicable):**	**SCSE1013, SCSE1224**
Credit hours:	**3**		

Course synopsis	This course provides students with fundamental and intermediate knowledge of data structures and algorithms using C++, emphasizing problem-solving, programming efficiency, and software development principles. The course covers essential topics in data structures, including abstract data types, arrays, linked lists, stacks, queues, trees, and file operations. Students will explore recursion, algorithm efficiency, and Big O analysis to evaluate performance. Sorting and searching algorithms, from basic to advanced techniques, are also emphasized, alongside practical implementations of data structures in C++. The course concludes with hands-on project demonstrations to consolidate understanding. This course adopts lectures, lab exercises, and project-based learning strengthen both theoretical understanding and practical programming skills. By the end of the course, students will be able to analyze and implement suitable data structures and algorithms to solve computational problems effectively. They will also demonstrate problem-solving, programming, and teamwork skills.			
Course coordinator (if applicable)	Ts. Dr. Johanna binti Ahmad			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Dr Ruhaida Samsudin			ruhaidah@utm.my
	Dr Fazliaty Edora Fadzli			fazliaty.edora@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO*	PLO **(MQF Cluster Code)	***Taxonomies and ****Graduate attributes	T&L methods	*****Assessment methods
CLO1	Apply suitable data structures and implement effective algorithms to solve programming problems.	PLO1 (C1)	C3 (SC1)	Lecture, active learning	Q,T, F
CLO2	Analyze programming problems and determine appropriate data structure concepts for effective implementation.	PLO2 (C2)	C4 (SC2)	Lab work	F,L, LT
CLO3	Develop medium-scale programs in a mini project by applying suitable data structure concepts to solve programming problems.	PLO2 (C2)	C4 (SC2)	Project-based learning	Asg, PR
CLO4	Evaluate and demonstrate effective teamwork in completing a group mini project.	PLO7 (C3E)	A3 A (IT3)	Project-based learning	PR

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.
*Up to 5 CLO
Refer ***Taxonomies of Learning and ****UTM's Graduate Attributes for measurement of outcomes achievement
*****T – Test; LT- Lab test; L- Lab exercise; Q – Quiz; HW – Homework; Asg – Assignment; PR – Project; Pr – Presentation; F – Final Exam etc.

****MQF Cluster Code**
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numercy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	Conducted through in-class activities
2	Independent Study	E-learning, MOOCs
3	Project-based learning	Conducted through case study assignment and mini projects. Tasks are given in sequential steps throughout the semester. Students in a group of 3 are given a case study that requires them to use data structure concepts to solve certain problems efficiently.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :
This course equips students with a range of transferable skills that extend beyond technical knowledge and can be applied in diverse academic, professional, and real-world contexts. Students develop strong problem-solving and analytical thinking abilities, supported by critical thinking and effective decision-making skills.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO* Compulsory	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1	Chapter 1: Introduction to Data Structures and Algorithms ●Introduction to Data Structure ●Types of Data Structure ●Algorithms ●Programming Principle	CLO1	2			1						2	5
Week 2	Chapter 2: Abstract Data Type, C++ Revision and Advanced File Operations ●Abstract Data Type Concepts ●C++ Revision ●C++ class ●Pass objects to functions, class as return value for functions ●Array of class and pointer to class ●Advanced File Operations ●File Operations ●Passing File Stream Objects to Functions ●Member Functions for Reading and Writing Files ●Binary Files ●Random-Access Files	CLO1, CLO2	2			1						2	5
Week 3	Chapter 3 & 4: Recursive and Algorithm Efficiency ●Recursive Principle and design ●Tracing Recursive Algorithms ●Algorithm Efficiency ●Analysis order of magnitude, Big O notation	CLO1, CLO2	2			1					2	2	7
Week 4	Chapter 5: Sorting ●Simple Sort •Bubble Sort •Selection Sort •Insertion Sort •Sorting Analysis	CLO2, CLO3	2			1					2	1	6
Week 5	Advanced Sort •Merge Sort •Quick Sort	CLO1, CLO2	2		1						1	1	5
Week 6	Chapter 6: Searching ●Sequential Search ●Binary Search	CLO1, CLO2	2		1							2	5
Week 7	Chapter 7: Linked List ●Pointer Concepts ●Introduction to Linked lists ●Linked lists operations ●Types of Linked List	CLO2, CLO3	2		1							2	5
Week 8	Linked List (continued) ●Linked List Implementations ●Declaring Nodes and Linked Lists class ●Insert Node, Delete Node, Find Node, Print Node	CLO2, CLO3	2		1						1	1	5
Week 9	Chapter 8: Stack ●Introduction to Stack ●Stack Operations ●push(), pop()	CLO2, CLO3	2			1					1	1	5
Week 10	Chapter 8: Stack (Continue) ●Stack implementations ●Stack Array ●Stack Linked List ●Stack Applications ●Infix, Postfix and Prefix	CLO2, CLO3	2			1					1	1	5
Week 11	Chapter 9: Queue ●Introduction to Queue ●Queue Implementations ●Array Implementation (Linear and Circular) ●Linked List Implementation (Linear and Circular Queue)	CLO1, CLO2	2			1					2	1	6
Week 12	Chapter 10: Tree •Introduction to Tree •Terms related to Tree concepts •Binary Search Tree	CLO1, CLO2	2			1					3		6
Week 13	Binary Search Tree Implementations •Declaring Tree node, Tree class •Create Node, Insert Node, Delete Node, Search Node •Tree Traversals	CLO1, CLO2	2			1					1	1	5
Week 14	Final Exam Revision (Searching, Linked List, Stack, Queue,Tree) Group Project Demo/Presentations	CLO1, CLO2	2			1					2	2	7
											SUB-TOTAL SLT :		77

	Continous Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Quiz	10	0.5			2.5	3
2	Lab Exercise	10	3		1	2	6
3	Assignment	10	3		2	2	7

4	Mini Project	10	3		2	2	7
5	Lab Test	15	3		2	2	7
6	Mid Term Test	15	3		2	2	7
					SUB-TOTAL SLT :		**37**

	Summative Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Final Exam	30	3			3	6
2							0
3							0
4							0
5							0
					SUB-TOTAL SLT :		**6**
					SLT for Assessment:		**43**
					GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	50.42
B	% SLT for Online & Independent Learning Component :	49.58
C	%SLT for Online Component:	20.83
D	% SLT for All Practical Component:	3.33
D1	% SLT for F2F Physical Practical Component:	3.33
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)

Computer lab that is equipped with programming software (eg: Dev C++, Microsoft VS Code), Draw UML Diagaram(eg: Enterprise Architect, Draw.io)

References (include required and further readings, and should be the most current)

Main references

Nor Bahiah, A., Dayang Norhayati, A., & Zalmiyah, Z. (2014). Data structures and algorithms (3rd ed.). Universiti Teknologi Malaysia. (Compulsory)

Additional references

Farrier, J. (2024). Data structures and algorithms with the C++ STL: A guide for modern C++ practitioners. Packt Publishing.
Malhotra, D., & Malhotra, N. (2024). Data structures and program design using C++: A self-teaching introduction to data structures and C++. Packt Publishing.
Carrano, F. M., & Henry, T. (2017). Data abstraction and problem solving with C++: Walls and mirrors (7th ed.). Pearson.Online

Course Website:

Utm E-learning at http://elearning.utm.my
Massive Open Online Learning Data Structures and Algorithms: https://www.openlearning.com/courses/SCSJ2013x

Other additional information (if applicable)

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

All students must adhere to the following to pass a computing course:
i.Pass both Continuous Assessment AND Final Assessment.
ii.Get at least minimum pass marks (40%) and pass grade (D+) following UTM regulation.

The passing mark for:
i.Continuous Assessment is 30% of continuous marks.
ii.Final Assessment is 20% of the final marks.

If a student does not achieve the passing mark for Continuous Assessment OR the passing mark for the Final Assessment, then the student's overall final mark will be set below the UTM passing mark.

Disclaimer:

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ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours

Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by: **Name:** **Signature:** **Date:**	**Certified by:** **Name:** Radziah Mohamad **Signature:** **Date:** 6/10/2025

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